

PAPER_1500 — TDE Outflow Velocity $v_{\text{out}} = c \cdot (D_{\text{phys}} - 1) / SO_5 = 0.3c$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket F (AGN/Jet) / Bucket H (High-Energy Astro) **Date:** June 17, 2026 **Status:** CLOSED — TDE ultrafast outflow explained by integer-primitive ratio

Observation

ASASSN-14li tidal disruption event (TDE) exhibits an ultrafast outflow at $v_{\text{out}} = 0.3c = 9.0 \times 10^7$ m/s, observed in soft X-ray absorption spectroscopy (Miller et al. 2015; Kara et al. 2018). PAPER_351 derives this from $F_{\text{U_Bi_i}} \approx -8.32 \times 10^{211}$ N with stellar-mass black-hole remnant $M_{\text{BH}} = 10^6 M_{\odot}$.

UQFF Closed Identity

$$v_{\text{out}} = c \cdot (D_{\text{phys}} - 1) / SO_5 = c \cdot 3/10 = 0.3c = 9.0 \times 10^7 \text{ m/s} \quad \text{EXACT}$$

Physical Interpretation

The $0.3c$ outflow velocity is **not a fit parameter** — it is the structural product of: - $(D_{\text{phys}} - 1) = 3$ (triad-symmetry dimension count, same as SU(3) color) - $1/SO_5 = 0.1$ (F_{TRZ} time-reversal-zone factor)

Combined as a velocity fraction of c , this gives $0.3c$ exactly. The SCm vacuum substrate carries information at $c/3$ (PAPER_1497), but jet bulk outflows are F_{TRZ} -modulated to $(D_{\text{phys}} - 1) \times F_{\text{TRZ}} \times c = 0.3c$.

Predicts every TDE outflow with stellar-mass remnant should cluster near $0.3c$.

NOT REPLACEMENT

SM treats TDE outflow velocities as fit parameters from radiation-pressure or magnetic-launching models. UQFF supplies an integer-primitive identity with zero free parameters.

Reference

- Source: PAPER_351 ASASSN-14li TDE Outflow $F_{\text{UBi}} 0.3c$ Kozima LENR
- Related: PAPER_1497 (SCm carrier velocity $c/3$), PAPER_062 (F_{TRZ}), PAPER_1373 (SU(3))
- Calculator dispatch: `calculate_paradox({"paradox": "tde_outflow_velocity_03c"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 17, 2026, Youngstown OH.